

Previous: [Terminal Input](#), Up: [Basic Input and Output](#) [[Contents](#)][[Index](#)]

14.1.3 Simple File I/O

The `save` and `load` commands allow data to be written to and read from disk files in various formats. The default format of files written by the `save` command can be controlled using the functions `save_default_options` and `save_precision`.

As an example the following code creates a 3-by-3 matrix and saves it to the file `'myfile.mat'`.

```
A = [ 1:3; 4:6; 7:9 ];
save myfile.mat A
```

Once one or more variables have been saved to a file, they can be read into memory using the `load` command.

```
load myfile.mat
A
-| A =
-|
-|    1    2    3
-|    4    5    6
-|    7    8    9
```

```
: save file
: save options file
: save options file v1 v2 ...
: save options file -struct STRUCT
: save options file -struct STRUCT f1 f2 ...
: save - v1 v2 ...
: str = save ("-", "v1", "v2", ...)
```

Save the named variables `v1`, `v2`, ..., in the file `file`.

If no variable names are listed, Octave saves all the variables in the current scope. Otherwise, full variable names or pattern syntax can be used to specify the variables to save. If the `-struct` modifier is used then the fields of the **scalar** struct are saved as if they were variables with the corresponding field names. The `-struct` option can be combined with specific field names `f1`, `f2`, ... to write only certain fields to the file.

The `save` command may also be invoked using the functional form

```
save ("-option1", ..., "file", "v1", ...)
```

where the `options`, `file`, and variable name arguments (`v1`, ...) must be specified as character strings.

Valid options for the `save` command are listed in the following table. Options that modify the output format override the format specified by `save_default_options`.

`-append`

Append to the file instead of overwriting.

`-ascii`

Save a matrix in a text file without a header or any other information. The matrix must be 2-D and only the real part of any complex value is written to the file. Numbers are stored in single-precision format and separated by spaces. Additional options for the `-ascii` format are

`-double`

Store numbers in double-precision format.

`-tabs`

Separate numbers with tabs.

`-binary`

Save the data in Octave's binary data format.

`-float-binary`

Save the data in Octave's binary data format using just single precision. Use this format **only** if you know that all the values to be saved can be represented in single precision.

`-hdf5`

Save the data in HDF5 format. (HDF5 is a free, portable, binary format developed by the National Center for Supercomputing Applications at the University of Illinois.) This format is only available if Octave was built with a link to the HDF5 libraries.

`-float-hdf5`

Save the data in HDF5 format using just single precision. Use this format **only** if you know that all the values to be saved can be represented in single precision.

`-text` (default)

Save the data in Octave's text data format. The [save_precision](#) function specifies the number of significant figures to use when saving data (default: 17). The header of the text data file can be configured with [save_header_format_string](#).

`-v7.3`

`-V7.3`

`-7.3`

Octave does **not** yet implement saving in MATLAB's v7.3 binary data format.

`-v7`

`-V7`

`-7`

`-mat7-binary`

Save the data in MATLAB's v7 binary data format.

`-v6`

`-V6`

```
-6
-mat
-mat-binary
```

Save the data in MATLAB's v6 binary data format.

```
-v4
-V4
-4
-mat4-binary
```

Save the data in MATLAB's v4 binary data format.

```
-zip
-z
```

Use the gzip algorithm to compress the file. This works on files that are compressed with gzip outside of Octave, and gzip can also be used to convert the files for backward compatibility. This option is only available if Octave was built with a link to the zlib libraries.

The list of variables to save may use wildcard patterns (glob patterns) containing the following special characters:

```
?
```

Match any single character.

```
*
```

Match zero or more characters.

```
[ list ]
```

Match the list of characters specified by *list*. If the first character is `!` or `^`, match all characters except those specified by *list*. For example, the pattern `[a-zA-Z]` will match all lower and uppercase alphabetic characters.

Wildcards may also be used in the field name specifications when using the `-struct` modifier (but not in the struct name itself).

Programming Notes: If called with the special filename `"-"` the data to be saved is returned as a string rather than writing it to an actual file.

When saving global variables the global status of the variable is also stored. If the variable is restored at a later time using `load`, it will be restored as a global variable. Global status is *not* preserved if using a MATLAB binary data file format or the `-ascii` format.

Example:

The command

```
save -binary data a b*
```

saves the variable `'a'` and all variables beginning with `'b'` to the file `data` in Octave's binary format.

See also: [load](#), [save_default_options](#), [save_header_format_string](#), [save_precision](#), [csvwrite](#), [dlmwrite](#), [fwrite](#).

There are three functions that modify the behavior of *save*.

```
: val = save_default_options ()
: old_val = save_default_options (new_val)
: old_val = save_default_options (new_val, "local")
```

Query or set the internal variable that specifies the default options for the *save* command, and defines the default format.

The default value is *"-text"* (Octave's own text-based file format). See the documentation of the *save* command for other choices.

When called from inside a function with the *"local"* option, the variable is changed locally for the function and any subroutines it calls. The original variable value is restored when exiting the function.

See also: [save](#), [save_header_format_string](#), [save_precision](#).

```
: val = save_precision ()
: old_val = save_precision (new_val)
: old_val = save_precision (new_val, "local")
```

Query or set the internal variable that specifies the number of digits to keep when saving data in text format.

The default value is 17 which is the minimum necessary for the lossless saving and restoring of IEEE-754 double values; For IEEE-754 single values the minimum value is 9. If file size is a concern, it is probably better to choose a binary format for saving data rather than to reduce the precision of the saved values.

When called from inside a function with the *"local"* option, the variable is changed locally for the function and any subroutines it calls. The original variable value is restored when exiting the function.

See also: [save_default_options](#).

```
: val = save_header_format_string ()
: old_val = save_header_format_string (new_val)
: old_val = save_header_format_string (new_val, "local")
```

Query or set the internal variable that specifies the format string used for the comment line written at the beginning of text-format data files saved by Octave.

The format string is passed to *strftime* and must begin with the character '#' and contain no newline characters. If the value of *save_header_format_string* is the empty string, the header comment is omitted from text-format data files. The default value is

```
"# Created by Octave VERSION, %a %b %d %H:%M:%S %Y %Z <USER@HOST>"
```

When called from inside a function with the *"local"* option, the variable is changed locally for the function and any subroutines it calls. The original variable value is restored when exiting the function.

See also: [strftime](#), [save_default_options](#).

```
: load file
: load options file
: load options file v1 v2 ...
: S = load ("options", "file", "v1", "v2", ...)
: load file options
: load file options v1 v2 ...
: S = load ("file", "options", "v1", "v2", ...)
```

Load the named variables *v1*, *v2*, ..., from the file *file*.

If no variables are specified then all variables found in the file will be loaded. Otherwise, full variable names or pattern syntax can be used to specify the variables to save. The format of the file is automatically detected but may be overridden by supplying the appropriate option.

The *load* command may also be invoked using the functional form

```
load ("-option1", ..., "file", "v1", ...)
```

where the *options*, *file*, and variable name arguments (*v1*, ...) must be specified as character strings.

Valid options for *load* are listed in the following table.

-ascii

Force Octave to assume the file contains columns of numbers in text format without any header or other information. Data in the file will be loaded as a single numeric matrix with the name of the variable derived from the name of the file.

-binary

Force Octave to assume the file is in Octave's binary format.

-hdf5

Force Octave to assume the file is in HDF5 format. (HDF5 is a free, portable binary format developed by the National Center for Supercomputing Applications at the University of Illinois.) Note that *load* is only designed to read HDF5 files that were created by Octave with *save*, and attempts to read other HDF5 files may fail or produce unpredictable results. The *-hdf5* option provides a limited ability to read files created using MATLAB's *-v7.3* option (which saves in HDF5 format) although many data types are not yet supported. This format is only available if Octave was built with a link to the HDF5 libraries.

-text

Force Octave to assume the file is in Octave's text format.

-v7.3

-V7.3

-7.3

Octave does **not** yet implement MATLAB's v7.3 binary data format. As the v7.3 format is an HDF5 based format, the *"-hdf5"* option may be used to attempt to open a v7.3 format file, although most non-numeric data types are not yet